

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

Grade 6 Mathematics

FIRST TERM EXAMINATION

SY 2026–2027 · MATATAG

50 points · Number & Algebra and Geometry (Tessellation)
Includes Table of Specifications and Answer Key for the teacher

First Term Examination

Name: _____ Section: _____ Date: _____ Score: _____/50

Read each item carefully. Write your answer on the blank. Show solutions for the problem-solving part.

I. Multiple Choice (1 pt each, 20 pts)

Write the letter of the correct answer.

1. What is $23.6 + 9.45$?
A. 32.05 B. 33.05 C. 33.5 D. 32.5
2. Subtract: $40 - 18.72$
A. 22.28 B. 21.28 C. 21.38 D. 22.38
3. Multiply: 5.4×0.3
A. 1.62 B. 16.2 C. 0.162 D. 1.26
4. Divide: $7.2 \div 0.8$
A. 0.9 B. 9 C. 90 D. 0.09
5. $\frac{2}{5} + \frac{1}{5} =$
A. $\frac{3}{10}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$
6. $\frac{3}{4} - \frac{1}{2} =$
A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{2}{4}$ D. $\frac{2}{3}$
7. $\frac{2}{3} \times \frac{3}{4} =$
A. $\frac{5}{7}$ B. $\frac{6}{7}$ C. $\frac{1}{2}$ D. $\frac{6}{12}$
8. $\frac{3}{4} \div \frac{1}{2} =$
A. $\frac{3}{8}$ B. $1\frac{1}{2}$ C. $\frac{2}{3}$ D. 2
9. How many $\frac{1}{4}$ are there in 3?
A. 7 B. 12 C. 3 D. 4
10. The GCF of 16 and 24 is
A. 4 B. 6 C. 8 D. 48
11. The LCM of 6 and 8 is
A. 24 B. 48 C. 2 D. 12
12. Which lists multiples of 9?

- A. 1,3,9 B. 9,18,27 C. 9,12,15 D. 3,6,9
13. Write $2 \times 2 \times 2 \times 2 \times 2$ in exponential form.
A. 2^5 B. 5^2 C. 2^4 D. 10
14. The value of 3^3 is
A. 9 B. 27 C. 6 D. 12
15. Simplify: $4 + 2 \times 5$
A. 30 B. 14 C. 20 D. 11
16. Simplify: $(9 - 5)^2$
A. 8 B. 16 C. 14 D. 4
17. In GEMDAS, which is done FIRST?
A. Addition B. Grouping C. Division D. Subtraction
18. Which shape tessellates by itself?
A. Regular pentagon B. Regular hexagon C. Circle D. Regular octagon
19. Angles meeting at a point in a tessellation add up to
A. 180° B. 90° C. 360° D. 270°
20. Which Filipino art best shows tessellation?
A. Kundiman B. Banig weaving C. Tinikling D. Sipa

II. Computation (2 pts each, 16 pts)

21. $18.5 \times 2.4 =$ _____
22. $96.6 \div 4.2 =$ _____
23. $3\frac{2}{3} + 1\frac{1}{2} =$ _____
24. $5\frac{1}{4} - 2\frac{3}{4} =$ _____
25. $2\frac{1}{2} \times 1\frac{3}{5} =$ _____
26. GCF and LCM of 12 and 30 = _____
27. Value of $5^2 + 4^3 =$ _____
28. Simplify: $2^3 + (12 - 4) \div 2 =$ _____

III. Problem Solving (show solutions, 14 pts)

29. (3 pts) Aling Nena bought 3.5 kg of chicken at ₱220.00/kg. How much did she pay, and what is her change from ₱1,000?

30. (3 pts) A pitcher has $2\frac{1}{2}$ L of juice. If $\frac{3}{4}$ L is poured out, how much remains?

31. (4 pts) Two bells ring — one every 15 minutes, the other every 20 minutes. They ring together at 8:00 a.m. At what time do they next ring together?

32. (4 pts) A ribbon $7\frac{1}{2}$ m long is cut into $\frac{3}{4}$ -m pieces. How many pieces can be made?

Table of Specifications & Answer Key

First Term Examination · 50 points

Table of Specifications

Competency (code)	Items	Pts	%
Operations on decimals (6NA-Ia-1, Ib-2)	1–4, 21, 22, 29	13	26%
Add/subtract fractions (6NA-Ic-3)	5, 6, 23, 24, 30	9	18%
Multiply/divide fractions (6NA-Id-4)	7, 8, 9, 25, 32	10	20%
GCF and LCM (6NA-Ie-5)	10, 11, 12, 26, 31	10	20%
Exponents & GEMDAS (6NA-If-6)	13–17, 27, 28	10	20%
Tessellation (6MG-Ig-7)	18, 19, 20	3	6%
Total	32 items	50	100%

Cognitive spread: Remembering/Understanding (Part I) $\approx$ 40%, Applying (Part II) $\approx$ 32%, Analyzing/Problem-solving (Part III) $\approx$ 28%.

Answer Key — Part I (Multiple Choice)

1	2	3	4	5	6	7	8	9	10
B	B	A	B	B	B	B	B	C	C
11	12	13	14	15	16	17	18	19	20
B	B	A	B	B	B	B	B	C	B

Note for #2: $40 - 18.72 = 21.28$ (B). **#15:** $(9-5)^2 = 4^2 = 16$ (B).

Answer Key — Part II (Computation)

21) 44.4 22) 23 23) $5\frac{1}{6}$ 24) $2\frac{1}{2}$ 25) 4 26) GCF = 6, LCM = 60 27) $25 + 64 = 89$ 28) $8 + 4 = 12$

Answer Key — Part III (Problem Solving)

29) $3.5 \times 220 = \text{₱}770.00$; change = $1000 - 770 = \text{₱}230.00$.

30) $2\frac{1}{2} - \frac{3}{4} = \frac{10}{4} - \frac{3}{4} = \frac{7}{4} = 1\frac{3}{4}$ L.

31) LCM(15,20) = 60 minutes $\rightarrow$ **9:00 a.m.**

32) $7\frac{1}{2} \div \frac{3}{4} = \frac{15}{2} \times \frac{4}{3} = 10 \rightarrow$ **10 pieces.**